

AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous ML discovery through pure evidence.

PERSISTENT WORLD

FRESH SCIENTIST

EVIDENCE FIRST

One Agent. Any ML Task. End to End.

A general-purpose research agent that runs the entire research loop autonomously.

01 task.md

What to solve.

Objectives • constraints • evaluation

EXAMPLE

Rank short videos for each user and interaction context.

KuaiRand-Pure • long_view • GAUC + nDCG@5

02 PERSONAL.md

How you want it solved.

Preferences • research style • priorities

EXAMPLE

Run on macOS / Apple Silicon with uv.

≤ 15 min / experiment • never print credentials

Autonomous by default. Supervisable by design.

More experiments. The same search basin.

LONG CONTEXT

Experience carries momentum.

A persistent conversation retains knowledge—and commitment to the current direction.

NAIVE RESTART

A fresh start loses progress.

Relearn the evaluator. Repeat failures.
Recover the strongest implementation.

How do we inherit experience and reopen the search?

Preserve the
research world.

Reset the researcher.

Pass evidence, failures, and code to a fresh Scientist.

Two scientific roles. One safe runtime.

Scientist

Hypothesize · code · test · interpret

One trajectory

META

Audit · preserve · hand off

Across trajectories

Runtime

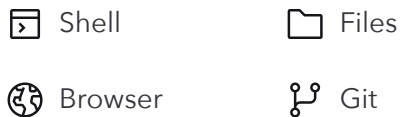
Isolation · cancellation · invariants

System lifetime

The Scientist chooses the science. META maintains the world. Runtime enforces mechanics, not scientific judgment.

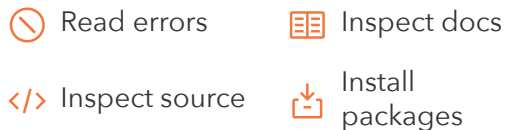
Research is an open-world task. We need an open-world coding agent.

01 OPEN ACTION SPACE



Use what exists

02 SELF-RECOVERY



Fix what breaks

03 RUNTIME EXTENSION



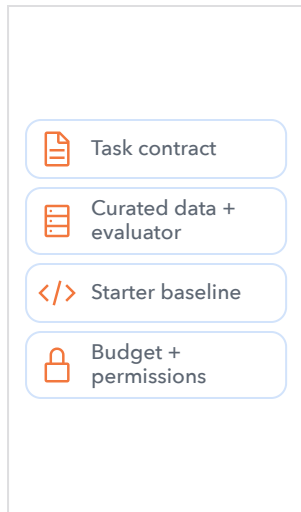
Add what's missing

We chose a coding agent—not a predefined workflow around known tools.

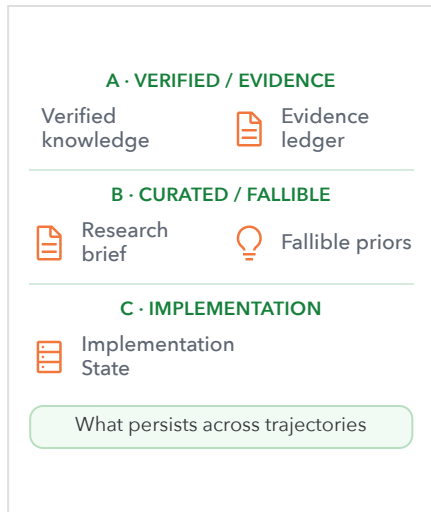
Missing capability → inspect → extend environment → retry

The persistence boundary is explicit.

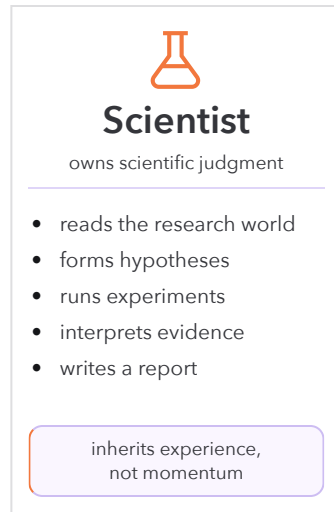
ENVIRONMENT



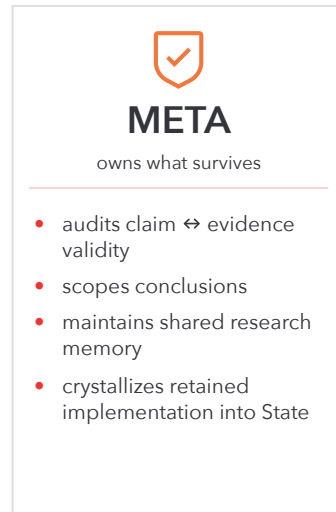
RESEARCH WORLD



FRESH SCIENTIST



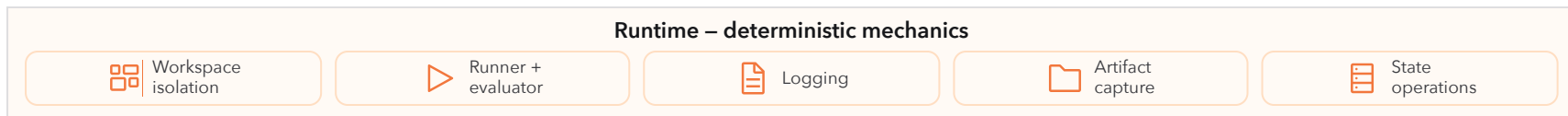
META REVIEW



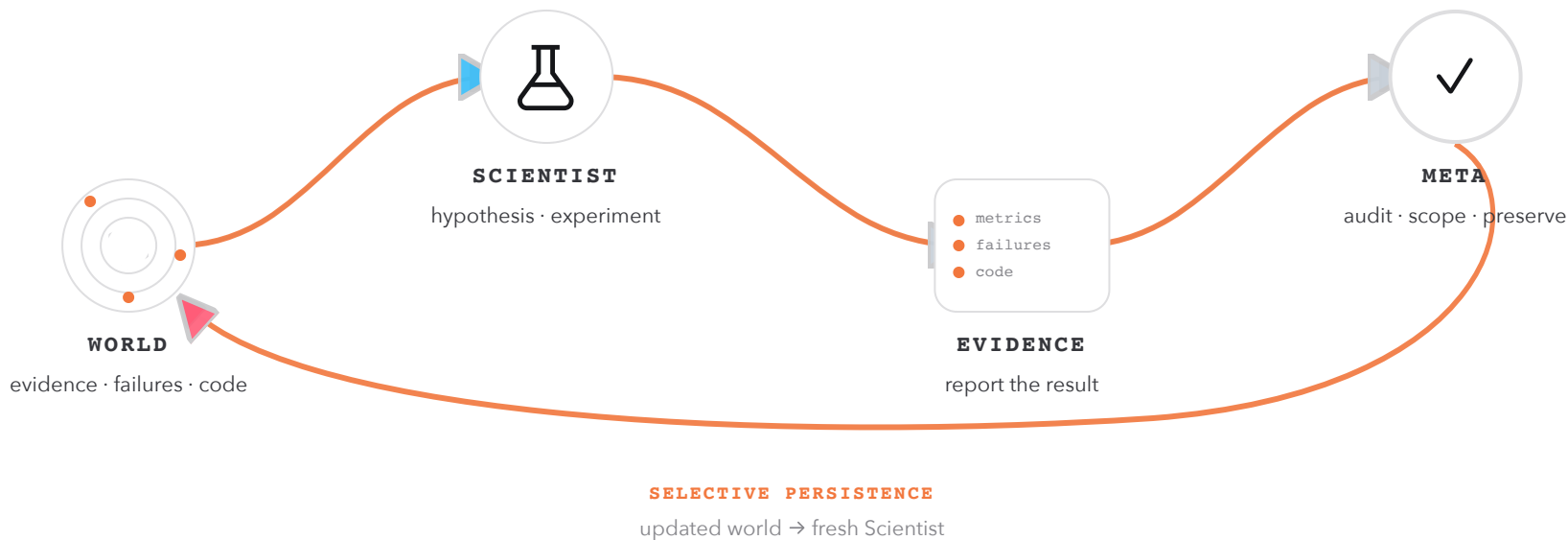
Selective persistence / trajectory handoff

audit → scope → preserve

Runtime – deterministic mechanics



Each cycle leaves a world to build on.



PERSISTS metrics · code · failures · original reports

RESTARTS an independent reasoning trajectory

Keep facts distinct from intuition.

research world

- └ evidence & provenance
- └ verified knowledge
- └ current research state
- └ fallible priors
- └ State + implementation

Built to be re-examined.

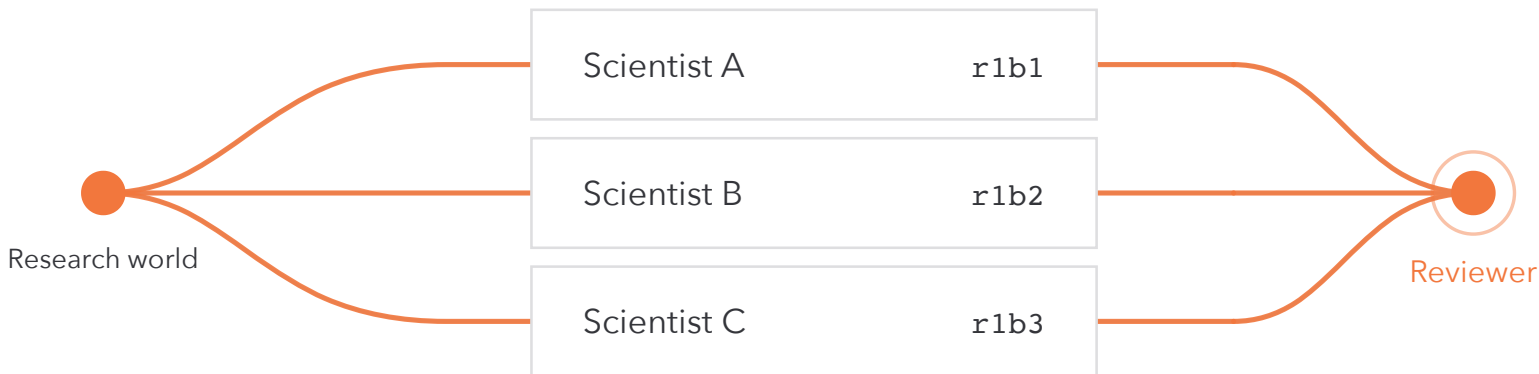
Facts have evidence.

Priors can be overturned.

Code can be picked up again.

Original reports remain available for audit.

One starting point. Independent answers.



Explore in isolated worktrees.

Review after completion.

Optional synthesis: one implementation parent + reference evidence from other branches.

Adoption is explicit via parallel-promote. Code and memory are not merged automatically.

Autonomous research. Measured gains.

PRIMARY / RESEARCH AGENT

0.6059363

+0.0043363 absolute improvement

METRIC	Official FM	Ours
GAUC	0.6674000	0.6728421
nDCG@5	0.5357000	0.5390304
Primary	0.6016000	0.6059363

Primary = mean(GAUC, nDCG@5). Public validation only; hidden-test scoring is organizer-controlled.

Final recipe: 46 categorical features × 8-seed FM ensemble · NumPy / CPU

We are NOT tokenmaxxing.

Better research decisions, not more resources.

~\$10

from first line of code to final result

coding • debugging • agent runs • all experiments

MODEL

One GPT Plus + One Gemini Pro for 7 days

Two consumer subscriptions covered the entire research loop.

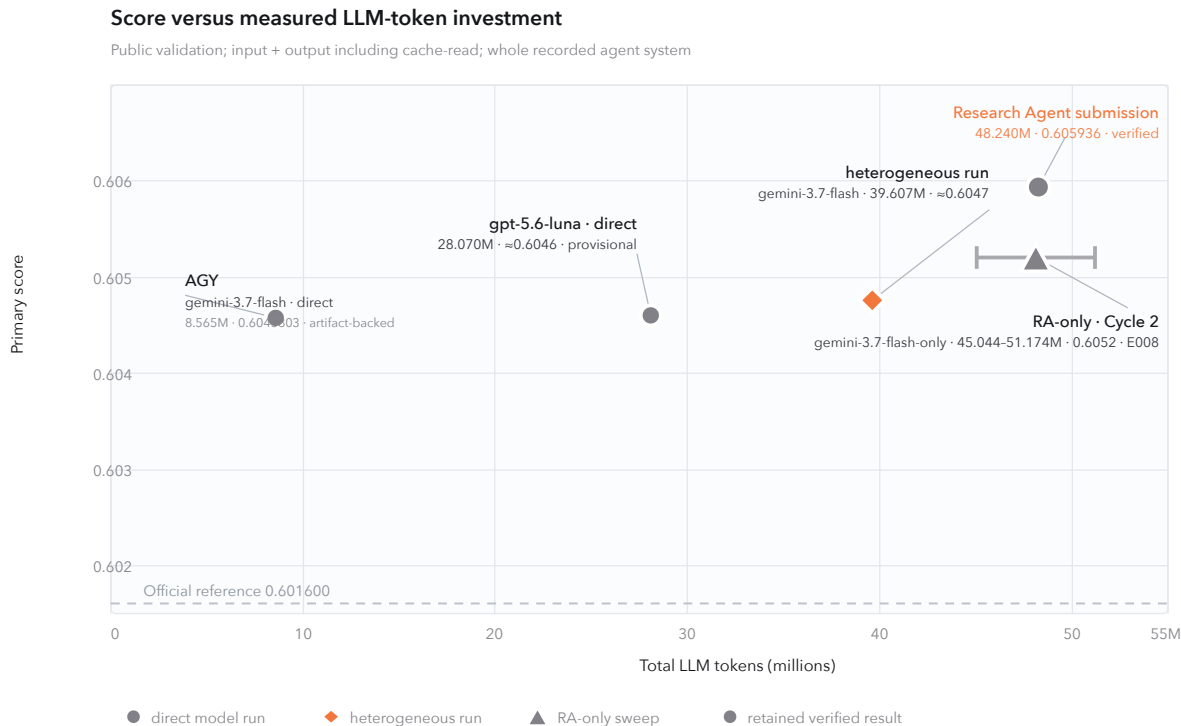
COMPUTE

1 × MacBook M2

All development and experiments ran on a single consumer laptop.

Performance came from the research loop – not a bigger budget.

The result stays connected to the evidence.



An audit trail from experiment to output.

4

Autonomous cycles

Within a 50-iteration cap

13

Named experiments

E001-E013 · 7 Full evaluations

0

Manual scientific interventions

After launch · 0 GPU-hours

170,588 prediction rows passed the unchanged Starter Kit alignment checker.

LLM tokens: 48.24M including cache reads; 4.02M excluding cache reads.

A retained state keeps the evidence inspectable.

The screenshot displays a 'Research Dashboard' with a 'RECORDED VALIDATION' status. The main content area features a 'Retained Validation Result' for checkpoint 'S004', which is highlighted as a 'RETAINED STATE'. This checkpoint is a 'SCORE-BEARING CHECKPOINT' and is described as an '8-seed ensemble of 46-field Factorization Machine rankers'. The dashboard provides key performance metrics: GAUC (0.672842), nDCG@5 (0.539030), Primary (0.605936), and Delta vs baseline (+0.004336). The evidence path is listed as 'system/evidence/cycle4-fm-rich46-ensemble8-full.json'. At the bottom, there are three navigation buttons: 'Research Intuition', 'Recent Activity', and 'Run Summary'.

Research Dashboard

RECORDED VALIDATION

★ Retained Validation Result

S004 · RETAINED STATE

SCORE-BEARING CHECKPOINT

S004

8-seed ensemble of 46-field Factorization Machine rankers incorporating leak-free historical user preferences (top tag, top tab, long-view rate bucket, interaction volume bucket, real-time tag/tab matches), secondary video tags, upload age, and per-seed Medium early stopping

GAUC
0.672842

nDCG@5
0.539030

Primary
0.605936

Delta vs baseline
+0.004336

Evidence · system/evidence/cycle4-fm-rich46-ensemble8-full.json

Research Intuition

Recent Activity

Run Summary

Inspect the retained state and the evidence behind it.

Retained validation checkpoint · evidence path · reproducible implementation.

Reproducible evidence. Open questions.

- | | | |
|----|--|--------------------|
| 01 | One task does not establish generality. | External validity |
| 02 | META, State, and reset are not isolated. | Causal attribution |
| 03 | Summaries can lose context and anchor. | Memory quality |
-

Next questions: broader tasks, targeted ablations, and less repeated execution.

Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD



Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture ·
integration



Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles ·
evidence review



Shilin Xu

DATA / METRICS

Data workflow · contracts ·
alignment



GE GAO

RUNTIME / RELIABILITY

Execution · testing ·
integration support



Jiran Li

EVIDENCE /
REPRODUCIBILITY

Telemetry ·
documentation ·
packaging

Keep the experience. Reopen the search.

```
# From the configured repo, run one autonomous cycle
$ ./scripts/research-agent step --cli codex \
  --target /absolute/path/to/project --allow-edits
```

github.com/zc6600/research-agent-kuairand ↗

Code · Technical report · Experiment ledger · Checked output

The run taught us more than the final score.

What survived the experiments – and what remains unsettled.

01 Diversity is a prior.

Keep `gemini-3.7-flash` as META while Scientist rotates across `gpt-5.6-sol`, `gemini-3.7-flash`, and `gpt-5.6-luna`. This widens priors; it does not prove the gain.

03 Failure can be evidence.

After evaluation, NumPy `float32` values broke serialization. The Scientist fixed the writer, reran, and kept the measurements.

02 A score is not a claim.

Audits found leakage and sampling mismatches in competing experiments. META can scope evidence; it is not a formal verifier.

04 Reset $\neq$ no anchor.

Fresh cognition reopens the search, but shared summaries can still omit context or anchor future work. Share evidence later than cognition.

OBSERVED, NOT PROVEN

Parallel/Synthesis reached public-validation Primary **0.6055536**; the canonical E001-E013 frontier remains **0.6059363**.

4 cycles

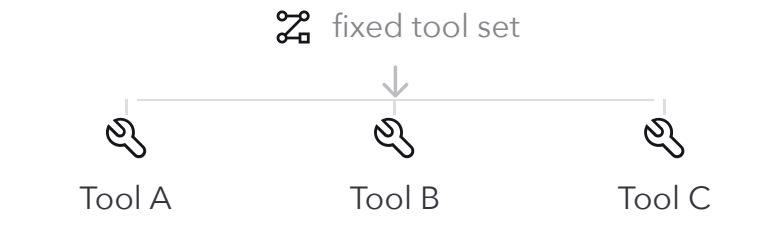
0 post-launch interventions

0 GPU-hours

The difference is not orchestration. It is recoverability.

LANGGRAPH-STYLE ORCHESTRATION

predefined graph



⊘ API fails



⊘ no recovery tool

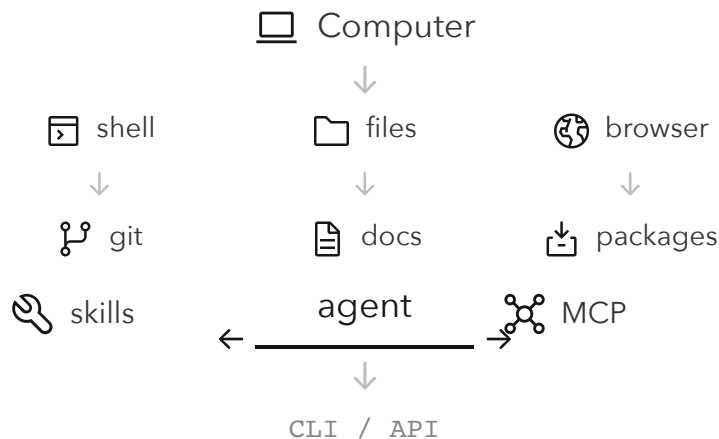


⊘ **DEAD END**

Capabilities are bounded by what developers anticipated.

CODING-AGENT HARNESS

open action space



fail → inspect → learn → modify environment → retry

The agent can acquire the missing capability at runtime.